Possession Without the Ball

Frame-level possession from team shape, ball proximity and a three-state decoder

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More than a Hack 2027 · Phase 1: Ball Possession · FC Barcelona vs Athletic Club (`BAR-ATH`)

Summary. The ball is missing on 41% of graded frames, and disproportionately at the moments that decide possession. We therefore read possession from the collective shape of each team as well as from the ball: the team in possession stretches the pitch wide and long, the team without it compacts. A ball-free gradient-boosted model of this behaviour is trained on *pseudo-labels* produced by a fixed rule from the tracking itself, and is combined with ball proximity, a dead-ball detector and the Smart Tagging inside a three-state hidden Markov model (`H/A/D`) whose Viterbi decoding charges a high price for a change of team. Because the ground truth may not be used for tuning, every setting was chosen with a *ball-masking validation*: the ball is hidden where the answer is known from the data, and the pipeline is scored on whether it still recovers it. No hand labels, ground truth or external data about this match were used.

1 Approach overview

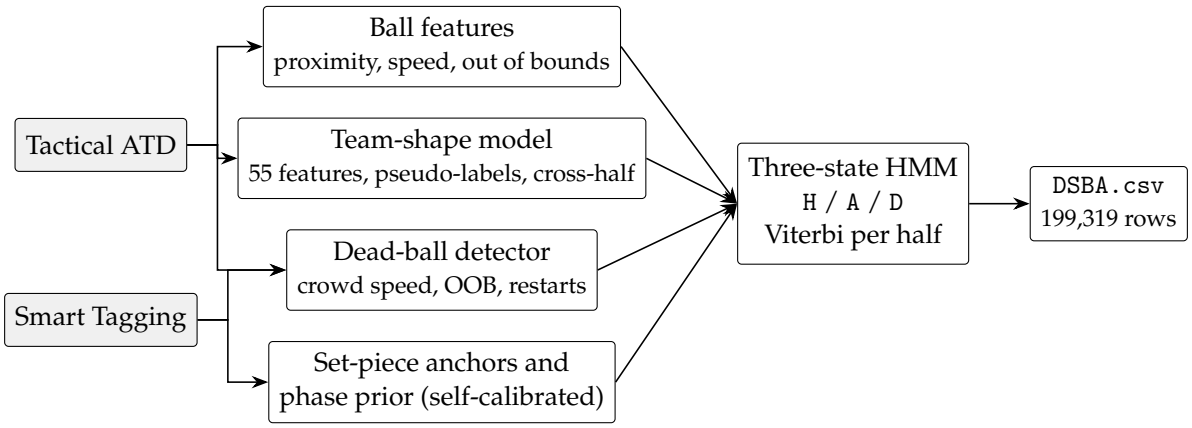


Figure 1: Pipeline. Each component produces per-frame log-odds; the decoder combines them.

Every component outputs evidence as per-frame log-odds rather than labels. Two separate questions are answered by separate evidence streams: *is the ball alive?* (dead-ball detector) and *if alive, which team has it?* (ball proximity, team shape, phase prior). The decoder then imposes the structure of a football match: possessions are long, changes of team are rare and must be sustained, and stoppages begin and end at identifiable moments.

2 Assets used and dropped

Asset	Use
Tactical ATD	All ball and player features: proximity, ball speed, out of bounds, team shape, collective speed.
Smart Tagging	Set pieces as restart anchors; phases as a weak team prior with learned direction; kickoff side.
Homography, both videos, broadcast ATD	**Not used.** The ATD is already in pitch coordinates; the broadcast feed has no alignment to the tactical timeline.
Public 5-minute ground truth	**Only** to verify frame alignment and file format. No setting was chosen with it.

3 Training data

Dataset	Source	Labelled by	Size
Pseudo-labels (§4.2)	Tactical ATD and Smart Tagging of this match	A fixed rule in `possession/pseudo.py`; no person	15,283 team frames; 16,759 dead frames

No other labelled data was used: no hand labels on any match, no ground truth (public or private), no pre-trained models. The unsupervised components (crowd-speed mixture, phase-tag direction) are fitted on this match's tracking without labels.

4 Method

4.1 Ball features (`possession/features.py`)

- **Cleaning.** Detections implying a speed above 40 m/s are discarded as implausible or teleporting; occlusions of at most 0.4 s are bridged.
- **Proximity log-odds.** With d_t^H and d_t^A the distance in metres from the ball to the nearest home and away player,

$$e_t \;=\; \gamma_t \cdot \mathrm{clip}\Big(d_t^A - d_t^H,\, -3,\, 3\Big)\,,$$

  where $\gamma_t \in [0, 1]$ is reduced when no player is within about 2.5 m and when the ball is in flight (above 7 m/s). A loose ball therefore carries no evidence, and possession stays with the last team, as `CHALLENGE.md` specifies.
- **Out of bounds.** The ball is observed more than 0.75 m outside the lines.
- **Collective speed.** Median player speed, smoothed over 2 s.

4.2 Pseudo-labels (`possession/pseudo.py`)

Frames whose answer is unambiguous from the data are selected by a fixed rule, never by a person.

- **Team:** the ball is observed and slow (below 5 m/s), a player is within 1 m of it, and no opponent is within 3 m. 15,283 frames (9,490 home, 5,793 away).
- **Dead:** the ball is observed out of bounds; a tagged set piece before its estimated restart; or the graded frames before the published second-half kickoff. 16,759 frames.

4.3 Team-shape model (`possession/shape.py`, `possession/shape_model.py`)

Features. 55 ball-free features per frame. Per team: centroid, width, length, spread, height of the deepest outfield player, most advanced player, players in the opponent half, speed, *push* (centroid velocity towards goal) and free space (mean distance to the nearest opponent). For each, the home-minus-away difference, its 1 s mean and its 2 s trend.

Orientation. All features are measured towards the team's own attacking goal. The attacking direction is inferred per half from the data (the deepest player sits at the goal the team defends), not hard-coded.

Model. Gradient-boosted trees (`HistGradientBoostingClassifier`) trained *cross-half*: the model applied to half 1 is trained only on half 2, and vice versa. Every prediction used downstream is therefore out of sample.

Held-out half	Accuracy vs pseudo-labels	AUC	Logistic regression AUC
1	0.864	0.934	0.906
2	0.876	0.940	0.913

Table 1: Cross-half performance of the shape model against pseudo-labels.

What it learned. Ranked by permutation importance (drop in AUC): (1) home-minus-away lateral spread; (2) home-minus-away distance from the centre line; (3) home-minus-away length, 1 s mean; (4) home speed; (5) away lateral spread; (6) home-minus-away depth spread; (7) the 2 s trend of the push towards goal. In football terms, *the team in possession is the wider one*, the coaching principle of making the pitch big with the ball. The trees add about 0.03 AUC over a linear model.

Entry into the decoder. The model's log-odds ℓ_t^{shape} enter with weight $0.2\, w_t$, where the gate w_t is 1 when proximity is uninformative and 0 when a player is clearly on the ball. Shape evidence fills the gaps; it does not overrule a visible ball.

4.4 Dead-ball detector (`possession/deadball.py`, `possession/tagging.py`)

Log-odds of *dead* versus *live*, summed per frame over four sources:

- **Ball observed out of bounds:** $+4$.
- **Collective speed:** a two-component Gaussian mixture fitted without labels on log crowd speed, with a slow component at 0.66 m/s and a fast component at 1.54 m/s.
- **Ball rolling inside the pitch / ball unseen:** -0.7 and $+0.15$. Calibrated on the tags: the ball is unseen on 35% of tagged-dead frames against 18% of tagged attacking-phase frames, so a missing ball is weak evidence of a stoppage.
- **Set pieces:** dead before the estimated restart, live immediately after.

Locating the restart. Aggregate measurements of the tag windows showed that kickoffs are windows of about 5 s, while every other set piece is a fixed 35 s window in which players accelerate only 20–30% of the way in. The restart is therefore estimated *per set piece* as the frame of the largest rise in collective speed within the window. The first version used the window midpoint, which marked about 10 s of live play after each set piece as dead; the fix raised the validation score (§6) from 0.697 to 0.747 without the shape model and from 0.715 to 0.780 with it.

4.5 Smart Tagging phase prior, with self-calibrated direction

Each phase code contributes at most ± 0.25 log-odds per frame towards its tagged team. The *sign is learned*, not assumed: on frames where proximity is confident, we measure how often the tagged team holds the ball. Every `DEFENDING ...` code and `DEFENSIVE TRANSITION` came out negative (agreement 0.11–0.38), meaning the tag names the team *without* the ball; `BUILD UP` and `PROGRESSION` came out at 0.86–0.89. No assumption about code semantics was required.

4.6 Decoder (`possession/decode.py`)

States $y_t \in \{\texttt{H}, \texttt{A}, \texttt{D}\}$ are decoded independently per half by Viterbi:

$$\hat{y}_{1:T} \;=\; \arg\max_{y_{1:T}} \sum_{t=1}^{T} \Big[\phi_t(y_t) \;-\; c_t(y_{t-1}, y_t) \Big],$$

where ϕ_t are the per-frame log-emissions and c_t the transition costs below.

Emissions. Team evidence (e_t, $0.2\, w_t\, \ell_t^{\text{shape}}$, phase prior) decides only *which* team; it never makes a live state more likely than `D`. That decision is left entirely to the dead-ball detector. Separating the two raised the validation score from 0.608 to 0.697 (measured before the restart fix).

Transition	Cost	Reduced where
H ↔ A	14	Never. A touch, a deflection or a lost aerial duel cannot pay for it.
live → dead	7	To 1.5 near an observed ball out of bounds.
dead → live	7	To 2 within 3 s of an estimated restart.

Table 2: Transition costs, in log-likelihood units.

Since proximity evidence is clipped at 3 per frame, a change of team requires at least $\lceil 14/3 \rceil = 5$ frames of saturated proximity for the other side, or a longer stretch of shape evidence. This is the implemented notion of *controlled* possession.

Half boundaries. Half 1 starts live, biased towards the tagged kickoff side. Half 2 is D until the published kickoff (frame 113,742, FAQ.md). **Clean-up.** Possessions shorter than 0.8 s and stoppages shorter than 1.2 s are absorbed into their neighbours.

5 Possession rules as implemented

- **Loose or contested ball:** no team evidence; the last team keeps possession.
- **Change of possession:** requires *sustained* control by the other side, as quantified by the switch cost above.
- **Restarts:** the set-piece anchor fixes the moment play resumes. The side comes from the tag where it names one, otherwise from proximity (throw-ins are tagged N/A).
- **Dead ball:** always D; no team is ever assigned to a dead ball.

6 Setting selection without ground truth: ball-masking validation

Since the ground truth may not be used for tuning, we built a test that needs none, implemented in possession/validate.py:

1. Take frames whose answer is known from the data (the pseudo-labels).
2. Hide the ball in 2–6 s blocks around them: 15% of graded frames, half centred on team frames and half on dead frames, with a fixed seed.
3. Re-run the full pipeline. The shape model, trained cross-half on unmasked data, has never seen the half it predicts.
4. Score *balanced recovery* $B = \frac{1}{2}\left(\mathrm{Acc}_{\mathrm{team}} + \mathrm{Acc}_{\mathrm{dead}}\right)$: correct team on hidden team frames and D on hidden dead frames.

Configuration	B	Team	Dead	Possessions
No shape model, v1 emissions	0.699	0.566	0.832	274
No shape model, team-only emissions	0.747	0.524	0.971	276
+ Team-shape model, weight 0.2 (chosen)	**0.780**	0.588	0.972	281
+ Team-shape model, weight 0.4	0.781	0.590	0.972	311
+ Dead-shape model, weight 0.05	0.761	0.628	0.894	282
+ Dead-shape model, weight 0.2	0.708	0.685	0.730	301

Table 3: Ball-masking validation. All rows include the per-set-piece restart estimate.

Reading the table.

- With the ball hidden, the shape model reduces wrong-team calls from 13.3% to about 5%.
- Weights 0.2 and 0.4 recover equally well, but 0.4 produces 30 more possessions. Because spurious possessions are penalised directly by spellF1, the conservative 0.2 was chosen.
- The remaining team-frame errors are D calls during live play with the ball hidden: a slow build-up

resembles a stoppage. This is the hardest case.

Tried and dropped.

- **Ball-free dead-ball shape model** (cross-half AUC 0.82–0.85): traded dead accuracy for team accuracy and lowered B at every weight tried.
- **Snapping boundaries to detected touches** (`--snap`): no ground-truth-free way to verify it; on the synthetic match the difference was within noise (0.453 vs 0.464 `spellF1`).
- **Abstaining with `X` mid-possession** (`--abstain`): the scorer forward-fills `X`, so `spellF1` is unchanged while `macroF1`, the tie-break, falls (by 0.06 on the synthetic match).

7 Result

- **Output:** 301 possessions with a mean duration of 14.8 s; 30% of graded frames are `D`.
- **Public 5 minutes:** `spellF1` 0.42 against 0.33 for the published floor `baseline_smooth50`, and `macroF1` 0.68 against 0.62. This is a correctness check only; 17 possessions cannot rank methods.

8 Limitations and next steps

- **Live versus dead with the ball hidden** is the weakest component: about 40% of hidden live frames are decoded as `D`. A stronger dead-ball model would need cleaner pseudo-labels for fouls and other untagged stoppages.
- **Fixed priors** (switch, stop and restart costs) were not tuned. The masking validation measures recovery, not over-switching, so tuning them against it would be unsafe.
- **The broadcast feed** could fill ball gaps if aligned to the tactical timeline; this was not attempted.
- **Pseudo-labels inherit the blind spots of the proximity rule** (balls in the air, crowded boxes). They were kept strict for this reason.

9 Reproduction

```
py -m venv .venv                                    # Python 3.12
.venv/Scripts/python -m pip install -r requirements-solution.txt
# unpack the data bundle into ./data
#   (data/tactical/BAR-ATH_tactical_atd.txt, ...)
.venv/Scripts/python run_pipeline.py
#   -> evaluation/results/DSBA.csv, then validates it
```

- One command, about 65 s. The first run parses and caches the tracking in `data/tactical/.cache/` (gitignored).
- **Deterministic.** Two runs, and a fresh environment built from the pinned requirements, produce an identical file.
  MD5: `97f61dfa580bb6f80156284c834b4873`
- `python run_validation.py` re-runs the settings study of §6.
- `python synthetic/make_synthetic.py` builds a synthetic match in the same file formats, for development only (§10).

10 Synthetic data and data handling

`synthetic/make_synthetic.py` simulates a full match in the bundle's exact file formats, with its own ground truth. It was used to build and debug the pipeline before the gated data arrived, and because the competition terms forbid sharing match data with cloud-hosted tools. It is not a model of the real match and was not used to choose any setting.

`data/`, `synthetic_data/` and `.cache/` are gitignored. The pipeline prints only counts, shares, fitted aggregates and model metrics, never tracking rows.